Activity 3.4 – Waveform Monitoring

1.Design Approach:

Designed a waveform monitoring module using Verilog HDL to capture important processor signals such as program counter, instruction, ALU result and write-back data. Implemented clock-based sampling.

2.Verification Steps:

- Created a separate testbench.
- Applied reset to initialize sampled outputs.
- Provided sample PC values.
- Tested instruction input changes.
- Verified ALU and write-back data sampling.
- Observed the outputs in the waveform.
- Simulated using ModelSim.

3.Results:

The testbench checks the sampled processor signals at every positive clock edge. PASS/FAIL messages are displayed.

4.Issues Faced:

Signal sampling must occur at the correct clock edge. Reset values must be initialized properly. All monitored signals must be connected correctly.

5.Conclusion:

Successfully designed and verified a processor waveform monitoring module using Verilog HDL.